

**RELAPSES**  
GPA: 50-70%  
MPA: ≥34%

**85-90%**  
inflamed sinus discharge  
tiny air serous fluid  
SUBGLOTTIC  
PROPTOSIS  
thrombosis warning  
VTE ↑

**TMP-SMX (sinus)**  
isolated only

**ANCA WARNING**  
20% ANCA NEG (GPA)  
20% ANCA POS (MPA)

**TRIPTYCH**  
MUST DO BIOPSY (lung)  
MUST DO BIOPSY (kidney)  
MUST DO ECHO!

**>1000**  
eosinophilia >1000/μL  
PINK EOSINOPHIL DOTS

**ASTHMA INHALER**  
GRANULOMAS ✓  
ASTHMA almost universal

**DIFFUSE ALVEOLAR HEMORRHAGE**  
COUGHING UP BLOOD  
PAUCI-IMMUNE

**NO ENT**  
NO NODULES  
VISIBLY DIFFERENT

**ALL PATIENTS!**  
leading cause of death  
Echocardiogram  
mononeuritis  
nasal allergy

**GRANULOMAS ✓**  
ASTHMA almost universal



## 1. GPA — c-ANCA/PR3

### WHAT YOU SEE

Left figure labeled GPA, PR3 90%, c-ANCA, building granulomas

### WHAT IT ENCODES

Granulomatosis with polyangiitis (formerly Wegener's) is a pauci-immune, granulomatous, necrotizing small-vessel vasculitis. It is c-ANCA / anti-PR3 (proteinase-3) positive in ~80-90% of generalized disease, lower in limited disease. The classic board triad is upper airway + lower airway (lung) + kidney (RPGN). Mnemonic: GPA = 'Granulomas, PR3, Airways.'

### BOARD TRAP

Vignette shows saddle nose + cavitary lung nodules + crescentic GN, and the answer offers 'MPO/p-ANCA' — wrong; PR3/c-ANCA is the GPA serology. MPO/p-ANCA points to MPA or EGPA, which lack granulomas and ENT destruction. The discriminator is granulomatous inflammation + upper-airway destruction = PR3.

## 2. GPA sinus / nose

### WHAT YOU SEE

Inflamed ear, nasal discharge, subglottic, proptosis around GPA

### WHAT IT ENCODES

GPA upper-airway disease: chronic refractory sinusitis, bloody nasal crusting, cartilage destruction producing saddle-nose deformity, subglottic stenosis (can cause stridor/airway compromise), retro-orbital pseudotumor with proptosis, and serous otitis media with conductive hearing loss. ENT involvement occurs in up to 90% and is often the presenting feature. Subglottic stenosis is treated locally (dilation/steroid injection) because it responds poorly to systemic immunosuppression.

### BOARD TRAP

ENT destruction with saddle nose can be mistaken for cocaine/levamisole-induced midline destructive lesions, which also give high-titer ANCA. The discriminator: ask about intranasal cocaine and check anti-human neutrophil elastase (HNE) antibodies, which are positive in cocaine-induced disease but not true GPA.

## 3. GPA lung nodules / cavities

### WHAT YOU SEE

A chest X-ray on the GPA wall showing rounded, partly hollowed-out lung nodules — the cavitating nodules ARE GPA's necrotizing granulomatous lung lesions (fixed/cavitary, unlike EGPA's fleeting infiltrates).

### WHAT IT ENCODES

GPA lung disease: bilateral cavitating nodules/masses and fixed infiltrates are characteristic (vs the fleeting/migratory infiltrates of EGPA). Diffuse alveolar hemorrhage from capillaritis can also occur. Biopsy shows necrotizing granulomatous inflammation with geographic necrosis.

### BOARD TRAP

Cavitary lung nodules in an immunosuppressed-appearing patient can be misread as TB, fungus, or septic emboli. The discriminator is the accompanying triad (sinus + RPGN) and PR3-ANCA — but you must still exclude infection with cultures/stains before immunosuppressing, since steroids worsen a missed infection.

## 4. GPA kidney — RPGN

### WHAT YOU SEE

An inflamed, angry-red kidney on the GPA panel — the fiery kidney IS pauci-immune crescentic (rapidly progressive) glomerulonephritis, the lesion that marks severe GPA.

### WHAT IT ENCODES

GPA renal disease is pauci-immune crescentic (rapidly progressive) glomerulonephritis: rising creatinine over days-weeks, active urinary sediment (dysmorphic RBCs, RBC casts, proteinuria). 'Pauci-immune' means immunofluorescence shows little/no immune-complex deposition, distinguishing it from immune-complex GN and anti-GBM (linear IgG). Renal involvement defines severe disease requiring aggressive induction.

### BOARD TRAP

RPGN with linear IgG on immunofluorescence and lung hemorrhage is Goodpasture/anti-GBM, NOT GPA (which is pauci-immune). The discriminator is the IF pattern: pauci-immune = ANCA vasculitis, linear = anti-GBM, granular/'lumpy-bumpy' = immune-complex. Note ~10-30% of anti-GBM patients are also MPO-ANCA positive (double-positive).

## 5. TMP-SMX (sinus)

### WHAT YOU SEE

TMP-SMX (sinus) tag bottom left

### WHAT IT ENCODES

TMP-SMX (trimethoprim-sulfamethoxazole) reduces relapses in localized/upper-airway GPA, likely by eradicating nasal *Staphylococcus aureus* carriage (a relapse driver). It simultaneously serves as *Pneumocystis jirovecii* (PCP) prophylaxis during induction immunosuppression. Standard prophylactic dosing is one DS tablet 3x/week or one SS tablet daily.

### BOARD TRAP

TMP-SMX is adjunctive for limited/sinus GPA and for PCP prophylaxis — it is NOT induction therapy for organ-threatening disease. Choosing TMP-SMX alone for a patient with RPGN or alveolar hemorrhage is wrong; that requires steroids + rituximab or cyclophosphamide.

## 6. MPA — p-ANCA/MPO

### WHAT YOU SEE

Center figure labeled MPA, MPO 75%, PAUCI-IMMUNE

### WHAT IT ENCODES

Microscopic polyangiitis is a pauci-immune necrotizing small-vessel vasculitis that is p-ANCA / anti-MPO (myeloperoxidase) positive in ~60-75%. Unlike GPA it has NO granulomas and NO destructive upper-airway disease. The dominant presentation is the pulmonary-renal syndrome: rapidly progressive GN plus diffuse alveolar hemorrhage. Mnemonic: MPA = 'MPO, Microscopic, no granulomas.'

### BOARD TRAP

A pulmonary-renal syndrome with p-ANCA but the stem says 'saddle nose deformity' — that detail makes it GPA, not MPA, because MPA never destroys cartilage. Conversely, choosing GPA for a pulmonary-renal syndrome with NO ENT disease and NO granulomas on biopsy is wrong; absence of granulomas + MPO = MPA.



## 7. MPA alveolar hemorrhage

### WHAT YOU SEE

Diffuse red lungs DIFFUSE ALVEOLAR HEMORRHAGE, coughing blood

### WHAT IT ENCODES

MPA causes diffuse alveolar hemorrhage (DAH) from pulmonary capillaritis: hemoptysis, dropping hemoglobin, diffuse ground-glass/alveolar infiltrates, and a rising DLCO (CO is taken up by intra-alveolar blood). This with RPGN constitutes the classic pulmonary-renal syndrome. DAH is life-threatening and may warrant plasma exchange in addition to steroids + cyclophosphamide/rituximab.

### BOARD TRAP

A rising (not falling) DLCO with diffuse infiltrates and hemoptysis is the buzzword for alveolar hemorrhage, not pneumonia or edema. Attributing it to heart failure/infection and withholding immunosuppression in a confirmed capillaritis is the trap. Differential for pulmonary-renal syndrome: MPA, GPA, and anti-GBM — separate them by ENT/granulomas and IF pattern.

## 8. MPA — no ENT / no nodules

### WHAT YOU SEE

GI vasculitis, NO ENT, NO NODULES, visibly different tags

### WHAT IT ENCODES

MPA spares the upper airway (no sinusitis, no saddle nose) and does not form cavitary nodules — those are GPA features. MPA is capillaritis, not granulomatous, so lung disease manifests as hemorrhage/fibrosis rather than cavitating masses. It can cause GI and skin (palpable purpura) involvement plus mononeuritis multiplex. MPO-ANCA-associated interstitial lung disease (fibrosis) may even precede vasculitis by years.

### BOARD TRAP

Long-standing pulmonary fibrosis later developing RPGN with MPO-ANCA is MPA, not idiopathic pulmonary fibrosis (IPF). Calling it IPF and missing the treatable vasculitis is the trap — always check ANCA in 'IPF' patients who develop renal disease or hemoptysis.

## 9. EGPA — asthma

### WHAT YOU SEE

Right figure with inhaler labeled EGPA, ASTHMA almost universal

### WHAT IT ENCODES

Eosinophilic granulomatosis with polyangiitis (Churg-Strauss) classically evolves through three phases: allergic (adult-onset asthma + allergic rhinitis), eosinophilic (peripheral/tissue eosinophilia, eosinophilic pneumonia/gastroenteritis), then vasculitic. Late-onset, often steroid-dependent asthma is nearly universal. ANCA (MPO/p-ANCA) is positive in only ~30-40%.

### BOARD TRAP

A negative ANCA does NOT exclude EGPA — it is ANCA-positive in a minority. Ruling out EGPA in an asthmatic with eosinophilia and mononeuritis because 'ANCA is negative' is wrong. ANCA-positive EGPA skews toward glomerulonephritis/neuropathy/purpura; ANCA-negative skews toward cardiac and lung disease.

## 10. EGPA eosinophilia >1000

### WHAT YOU SEE

>1000 eosinophilia >1000/ $\mu$ L, pink eosinophil dots

### WHAT IT ENCODES

EGPA features marked peripheral eosinophilia (typically >1500/ $\mu$ L or >10% of WBCs) plus eosinophilic tissue infiltration. Biopsy shows eosinophil-rich necrotizing granulomas and small-vessel vasculitis. Eosinophil count tracks disease activity and falls rapidly with steroids.

### BOARD TRAP

Hypereosinophilia plus cardiac/lung infiltrates can also be idiopathic hypereosinophilic syndrome (HES) or eosinophilic leukemia. The discriminator for EGPA is the asthma + vasculitis (neuropathy, purpura, glomerulonephritis) backdrop; pure HES lacks vasculitis and may carry the FIP1L1-PDGFR $\alpha$  fusion (treat with imatinib, not steroids alone).

## 11. EGPA cardiac — leading death

### WHAT YOU SEE

Heart, echocardiogram, leading cause of death tag

### WHAT IT ENCODES

Cardiac involvement (eosinophilic myocarditis, cardiomyopathy, coronary vasculitis, pericardial effusion) is the LEADING cause of death in EGPA and is more common in ANCA-negative disease. Every EGPA patient needs an echocardiogram (and often cardiac MRI/troponin) at diagnosis. Cardiac disease upgrades the patient to severe/poor-prognosis status (a Five-Factor Score item) warranting cyclophosphamide-based induction.

### BOARD TRAP

In EGPA mortality is driven by the HEART, not the kidney (renal disease is less severe than in GPA/MPA). Picking 'renal failure' as the leading cause of death in EGPA is the trap — screen and treat cardiac involvement aggressively.

## 12. EGPA mono/nasal/allergy

### WHAT YOU SEE

Mononeuritis, nasal allergy tags right side

### WHAT IT ENCODES

EGPA neurologic disease: mononeuritis multiplex (e.g., foot drop, wrist drop) from vasculitic ischemia of the vasa nervorum is very common and is a severe-disease (Five-Factor) feature. Upper airway involvement is allergic (rhinitis, nasal polyps, sinusitis) rather than destructive — so no saddle nose. Asymmetric, painful, stepwise sensorimotor deficits are the classic exam pattern.

### BOARD TRAP

Mononeuritis multiplex is non-specific (also PAN, MPA, GPA, RA, diabetes). The discriminator pointing to EGPA is the asthma + hypereosinophilia context. Attributing asymmetric neuropathy in an asthmatic eosinophilic patient to 'diabetic neuropathy' misses a treatable vasculitis.



### 13. EGPA — mepo/benra

#### WHAT YOU SEE

MEPO, BENRA, GCs tags bottom right

#### WHAT IT ENCODES

EGPA treatment: glucocorticoids are the backbone for non-severe disease; severe disease (cardiac, GI, renal, severe neuropathy) needs cyclophosphamide or rituximab induction. Mepolizumab (anti-IL-5) is approved for relapsing/refractory non-severe EGPA and is strongly steroid-sparing; benralizumab (anti-IL-5 receptor) is an alternative. These biologics target the eosinophilic, asthma-predominant phenotype.

#### BOARD TRAP

Mepolizumab/benralizumab control the eosinophilic/asthmatic burden but are NOT adequate sole induction for organ-/life-threatening vasculitis (active myocarditis, RPGN). Choosing mepolizumab alone for severe vasculitic EGPA is wrong — that needs steroids + cyclophosphamide or rituximab.

### 14. Triptych — must biopsy/echo

#### WHAT YOU SEE

TRIPTYCH panel: MUST DO BIOPSY (lung), BIOPSY (kidney), ECHO!

#### WHAT IT ENCODES

Diagnostic workflow for AAV: confirm with tissue biopsy where feasible — lung (necrotizing granuloma in GPA/EGPA vs capillaritis in MPA) and kidney (pauci-immune crescentic GN with negative/scant immunofluorescence). For EGPA specifically, obtain an echocardiogram to screen for the cardiac involvement that drives mortality. Combine biopsy with ANCA serology rather than relying on either alone.

#### BOARD TRAP

Do not anchor on ANCA serology alone — biopsy confirms pauci-immune necrotizing vasculitis and excludes mimics. Treating presumptively without tissue (when tissue is obtainable) risks immunosuppressing an infection or malignancy mimicking vasculitis.

### 15. ANCA-negative warning

#### WHAT YOU SEE

ANCA WARNING: 20% ANCA NEG (GPA) banner

#### WHAT IT ENCODES

Roughly 10-20% of GPA (especially limited/localized disease confined to the ENT or eye) is ANCA-negative, and ~60% of EGPA is ANCA-negative. ANCA testing should pair indirect immunofluorescence (c-/p-ANCA pattern) with antigen-specific ELISA (PR3/MPO) for accuracy. A negative ANCA in a compatible clinical picture mandates biopsy rather than exclusion of vasculitis.

#### BOARD TRAP

The classic trap: 'ANCA is negative, therefore not vasculitis.' Wrong — diagnosis is clinical + histologic. Excluding GPA in a patient with saddle nose, cavitary nodules, and RPGN solely because ANCA is negative leads to a missed, organ-threatening diagnosis.